

ELECTRICITY -03

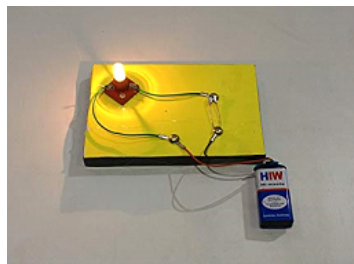
Class 10 - Science

Section A

1. **Read the text carefully and answer the questions:**

[4]

How do we express electric current? Electric current is defined by the amount of charge flowing through a particular area in unit time. In other words, it is the rate of flow of electric charges. In circuits using metallic wires, electrons constitute the flow of charges. However, electrons were not known at the time when the phenomenon of electricity was first observed. So, electric current was considered to be the flow of positive charges and the direction of flow of positive charges was taken to be the direction of electric current. Conventionally, in an electric circuit, the direction of electric current is taken as opposite to the direction of the flow of electrons, which are negative charges.

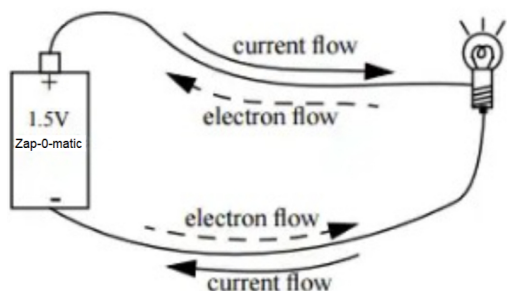


- If a net charge Q , flows across any cross-section of a conductor in time t , then the current I , through the cross-section is given by which formula?
- What is the SI unit of electric charge? It is equivalent to how many numbers of electrons?
- The electric current is expressed in which unit? Define the unit used to measure electric current.
- If the amount of charge passing through the cell in 4 seconds is 12 C then find the current supplied by a cell.

2. **Read the text carefully and answer the questions:**

[4]

The rate of flow of charge is called electric current. The SI unit of electric current is Ampere (A). The direction of flow of current is always opposite to the direction of flow of electrons in the current.



The electric potential is defined as the amount of work done in bringing a unit-positive test charge from infinity to a point in the electric field. The amount of work done in bringing a unit positive test charge from one point to another point in an electric field is defined as potential difference.

$$V_{AB} = V_B - V_A = \frac{W_{BA}}{q}$$

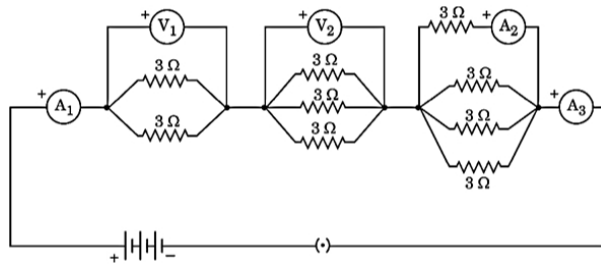
The SI unit of potential and potential difference is volt.

- Write the formula of voltage in terms of work done, current, time and charge.
- What is the number of electrons flowing per second in a conductor if 1 A current is passing through it?
- What would be the potential difference between the two terminals of a battery, if 100 joules of work is required to transfer 20 coulombs of charge from one terminal of the battery to other?
- The 2 C of charge is flowing through a conductor in 100 ms, then what would be the current in the circuit?

3. **Read the text carefully and answer the questions:**

[4]

Consider the following electrical circuit diagram in which nine identical resistors of $3\ \Omega$ each are connected as shown. If the reading of the ammeter A_1 is 1 ampere, answer the following questions:

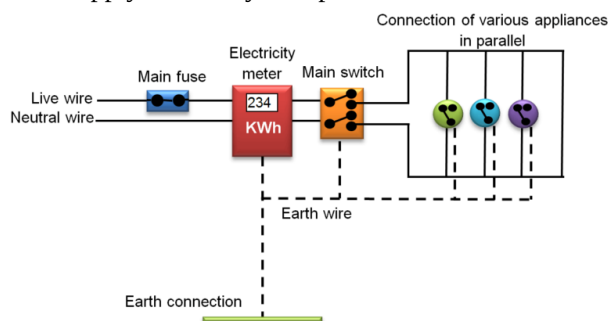


- What is the relationship between the readings of A_1 and A_3 ? Give reasons for your answer.
- What is the relationship between the readings of A_2 and A_3 ?
- Determine the reading of the voltmeter V_1 .
- Find the total resistance of the circuit.

4. **Read the text carefully and answer the questions:**

[4]

In our homes, either the overhead electric poles or underground cables support the power supply flowing through the mains supply. One of the wires in this supply is covered with insulation in the colour red, and another wire colored black. At the meter board, these wires pass into an electric meter through the main fuse. The main switch, live wire, and the neutral wire are in connection to the line wires in our homes; these wires then supply electricity to separate electric circuits within the house.



- What is the colour of the live wire?
- Where is the fuse placed in the electric supply in the above-given figure?
- What is the commercial unit of the power supply?
- What is the role of the fuse in series with any electrical appliance in an electric circuit?

5. **Read the text carefully and answer the questions:**

[4]

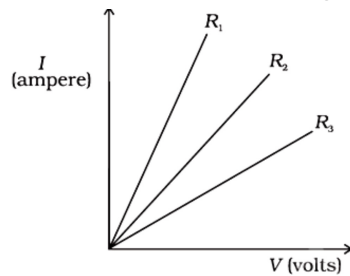
In 1827, a German physicist Georg Simon Ohm (1787-1854) found out the relationship between the current I , flowing in metallic wire and the potential difference across its terminals. He stated that the electric current flowing through a metallic wire is directly proportional to the potential difference V , across its ends provided its temperature remains the same.

The resistance of a circuit is defined as the ratio between the voltage applied to the current flowing through it.

Rearranging the above relation,

$$R = \frac{V}{I}$$

Electric charge flows easily through some materials than others. The electrical resistance measures how much the flow of this electric charge is restricted within the circuit.

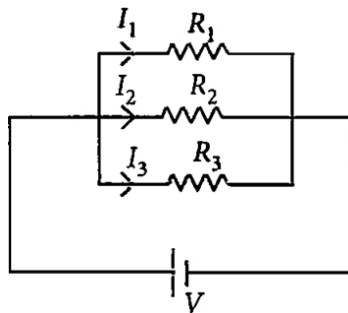


- What is the unit of electrical resistance?
- Define Ohm's law.
- From graph which resistance have high resistance?
- What does the slope of V-I graph at any point represent?

6. **Read the text carefully and answer the questions:**

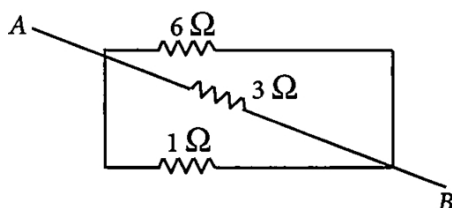
[4]

If two or more resistances are connected in such a way that the same potential difference gets applied to each of them, then they are said to be connected in parallel.

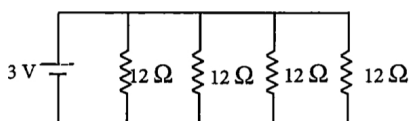


The current flowing through the two resistances in parallel is, however, not the same. When we have two or more resistances joined in parallel to one another, then the same current gets additional paths to flow and the overall resistance decreases.

- Three resistances, $2\ \Omega$, $6\ \Omega$ and $8\ \Omega$ are connected in parallel, then what will be the equivalent resistance?
- A wire of resistance $12\ \Omega$ is cut into three equal pieces and then twisted their ends together, then what will be the equivalent resistance?
- Three resistances are connected as shown. Calculate the equivalent resistance between A and B?



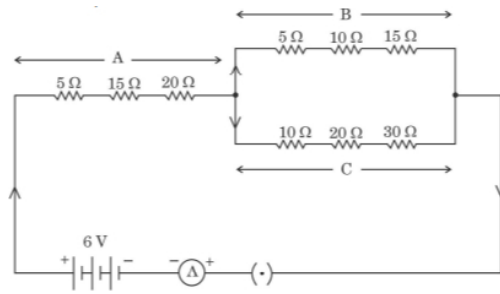
- Find the current in each resistance.



7. **Read the text carefully and answer the questions:**

[4]

Study the following electric circuit in which the resistors are arranged in three arms A , B and C:

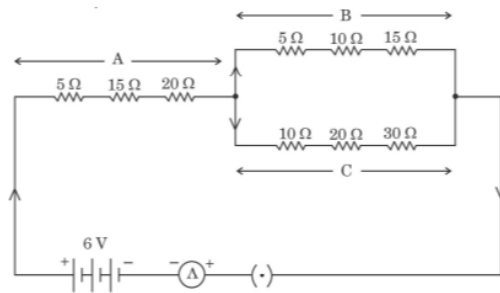


- Find the equivalent resistance of arm B.
- Calculate the equivalent resistance of the parallel combination of the arms B and C.
- Determine the current that flows through the ammeter.
- Determine the current that flows in the ammeter when the arm B is withdrawn from the circuit.

8. **Read the text carefully and answer the questions:**

[4]

Study the following electric circuit in which the resistors are arranged in three arms A, B and C:

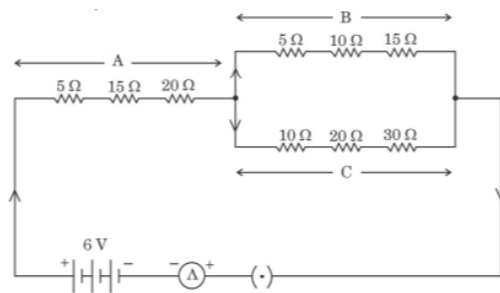


- Find the equivalent resistance of arm A.
- Calculate the equivalent resistance of the parallel combination of the arms B and C.
- Determine the current that flows through the ammeter.
- Determine the current that flows in the ammeter when the arm B is withdrawn from the circuit.

9. **Read the text carefully and answer the questions:**

[4]

Study the following electric circuit in which the resistors are arranged in three arms A, B and C :

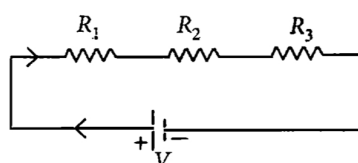


- Find the equivalent resistance of arm C.
- Calculate the equivalent resistance of the parallel combination of the arms B and C.
- Determine the current that flows through the ammeter.
- Determine the current that flows in the ammeter when the arm B is withdrawn from the circuit.

10. **Read the text carefully and answer the questions:**

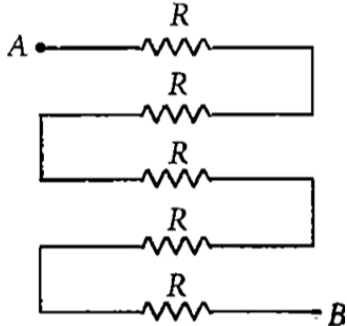
[4]

Two or more resistances are connected in series or in parallel or both, depending upon whether we want to increase or decrease the circuit resistance.

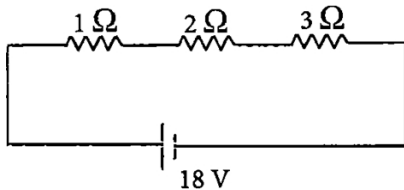


The two or more resistances are said to be connected in series if the current flowing through each resistor is the same.

- When the three resistors each of resistance R ohm are connected in series then what will be the equivalent resistance?
- There is a wire of length 20 cm and having resistance $20\ \Omega$ cut into 4 equal pieces and then joined in series. What is equivalent resistance?
- In the following circuit, find the equivalent resistance between A and B ($R = 2\ \Omega$)



- In the given circuit, what is the current in each resistor?



11. **Read the text carefully and answer the questions:**

[4]

The heating effect of current is obtained by transformation of electrical energy into heat energy. Just as mechanical energy used to overcome friction is covered into heat, in the same way, electrical energy is converted into heat energy when an electric current flows through a resistance wire. The heat produced in a conductor, when a current flows through it is found to depend directly on (a) strength of current (b) resistance of the conductor (c) time for which the current flows.

The mathematical expression is given by $H = I^2 R t$.

The electrical fuse, electrical heater, electric iron, electric geyser etc. all are based on the heating effect of current.

- What are the properties of heating element?
- What are the properties of electric fuse?
- When the current is doubled in a heating device and time is halved, what will be the heat energy produced?
- A fuse wire melts at 5 A. It is desired that the fuse wire of same material melt at 10 A. Find the new radius of the wire?

12. **Read the text carefully and answer the questions:**

[4]

The electrical energy consumed by an electrical appliance is given by the product of its power rating and the time for which it is used. The SI unit of electrical energy is Joule. Actually, Joule represents a very small quantity of energy and therefore it is inconvenient to use where a large quantity of energy is involved. So for commercial purposes, we use a bigger unit of electrical energy which is called kilowatt-hour. 1 kilowatt-hour is equal to 3.6×10^6 joules of electrical energy.

- The energy dissipated by the heater is E . When the time of operating the heater is doubled, what would be the energy dissipated?
- The power of a lamp is 60 W. What will be the energy consumed in 1 minute?
- The electrical refrigerator rated 400 W operates 8 hours a day. The cost of electrical energy is ₹5 per kWh.

Find the cost of running the refrigerator for one day.

- (d) Calculate the energy transformed by a 5 A current flowing through a resistor of 2Ω for 30 minutes.

13. **Read the text carefully and answer the questions:**

[4]

We know that a battery or a cell is a source of electrical energy. The chemical reaction within the cell generates the potential difference between its two terminals that sets the electrons in motion to flow the current through a resistor or a system of resistors connected to the battery. To maintain the current, the source has to keep expanding its energy. Where does this energy go? A part of the source energy in maintaining the current may be consumed for useful work (like in rotating the blades of an electric fan). The rest of the source energy may be expended in heat to raise the temperature of the gadget. We often observe this in our everyday life. For example, an electric fan becomes warm if used continuously for a long time, etc. On the other hand, if the electric circuit is purely resistive, that is, a configuration of resistors only connected to a battery; the source energy continually gets dissipated entirely in the form of heat. This is known as the heating effect of electric current. This effect is utilized in devices such as an electric heater, electric iron, etc.



- (a) Explain Joule's heating law.
- (b) In practical situations, when an electric appliance is connected to a known voltage source, then how does the heating effect of electric current can be calculated?
- (c) Write the relation between heat energy produced in a conductor when a potential difference V is applied across its terminals and a current I flows through for t .
- (d) Two identical wires one of nichrome and the other of copper are connected in series and a current (I) is passed through them. State the change observed in the temperatures of the two wires.

Section B

14. The unit for expressing electric power is:

[1]

- | | |
|------------|----------|
| a) Coulomb | b) Joule |
| c) Watt | d) Volt |

15. In a household, 250 unit electrical energy is consumed in particular months. In the standard unit, it means that the energy consumed is:

[1]

- | | |
|-------------|-------------|
| a) 250 KW/h | b) 250 KJ |
| c) 250 kWh | d) 500 kJ/h |

16. What current will be taken by a 920 W appliance if the supply voltage is 230 V?

[1]

- | | |
|------------------|--------|
| a) 4 A | b) 9 A |
| c) None of these | d) 1 A |

17. Unit of electric power may also be expressed as

[1]

- | | |
|------------------|----------------|
| a) joule second | b) volt ampere |
| c) kilowatt hour | d) watt second |

18. If the potential difference between the end of a wire of fixed resistance is doubled, by how much does the

[1]

electric power increase?

- a) $\frac{4V}{R}$ b) $\frac{3V}{R}$
c) $\frac{6V}{R}$ d) None of these

19. When an electric lamp is connected to 12 V battery, it draws a current of 0.5A. The power of the lamp is: **[1]**
a) 0.5W
b) 24W
c) 12W
d) 6W
 20. An electric bulb is rated 220 V and 100 W. When it is operated on 110 V, the power consumed will be- **[1]**
a) 100 W
b) 50 W
c) 25 W
d) 75 W
 21. A 2 kW heater, a 200 watt TV and three 100 watt lamps are all switched on from 6⁻¹⁰ PM. What is the total cost at Rupees 5.50 per kWh? **[1]**
a) 55
b) None of these
c) 56
d) 67
 22. Two conducting wires of the same material and of equal lengths and equal diameters are first connected in series and then parallel in a circuit across the same potential difference. The ratio of heat produced in series and parallel combinations would be- **[1]**
a) 1 : 2
b) 4 : 1
c) 2 : 1
d) 1 : 4
 23. What is the maximum power in kilowatts of the appliance that can be connected safely to a 13A, 230V mains socket? **[1]**
a) 2.95 kW
b) None of these
c) 2.99 kW
d) 2.97 kW
 24. In an electrical circuit three incandescent bulbs A, B and C of rating 40 W, 60 W and 100 W respectively are connected in parallel to an electric source. Which of the following is likely to happen regarding their brightness? **[1]**
a) Brightness of bulb B will be more than that of A
b) Brightness of bulb A will be the maximum
c) Brightness of all the bulbs will be the same
d) Brightness of bulb C will be less than that of B
 25. Which of the following is likely to be the correct wattage for an electric iron used in our homes? **[1]**
a) 60 W
b) 2000 W
c) 850 W
d) 250 W
 26. An electric bulb rated 220 V, 60 W is working at full efficiency. Another identical bulb is connected in the same circuit having power supply of 220 V. **[1]**
i. If both the bulbs are connected in series then the total power consumption will be 60 W.
ii. If only one bulb is connected then the total power consumption will be 30 W.
iii. If the both bulbs are connected in parallel then the total power consumption will be 120 W.

a) Only (iii)

b) Only (ii) and (iii)

c) Only (i) and (ii)

d) Only (i)

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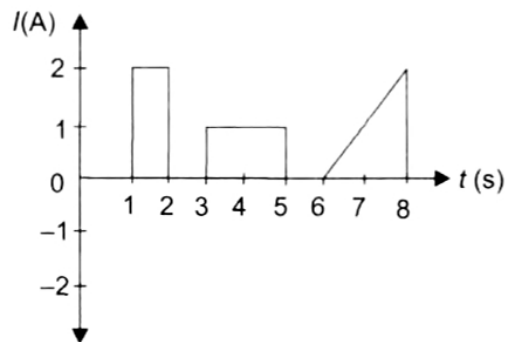
a) $\frac{l_1}{l_2} = \frac{2}{3}$

b) $\frac{l_1}{l_2} = \frac{3}{5}$

c) $\frac{l_1}{l_2} = \frac{1}{4}$

d) $\frac{l_1}{l_2} = \frac{1}{2}$

35. The plot represents the flow of current through a wire at three different time intervals. The ratio of charges flowing through the wire at different intervals is [1]



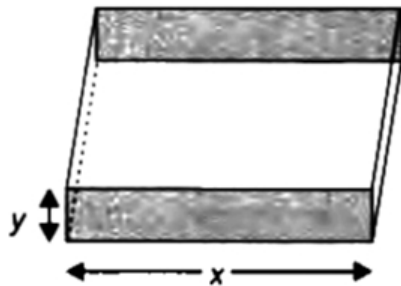
a) 1 : 2 : 3

b) 3 : 2 : 2

c) 2 : 3 : 3

d) 1 : 1 : 1

36. Consider a thin square sheet of side x and thickness y made of a material of resistivity ρ . The resistance between two opposite faces, shown by the shaded areas in the figure is [1]



a) Directly proportional to x

b) Independent of x

c) Independent of y

d) Directly proportional to y